

Jenkins and build triggers

Lab 6

Improve the execution of the Pipeline of Lab 5.

You need to make following improvements:

1. Add a git hook that triggers build automatically when a change is committed.
2. Make test scripts installation independent. All local installation related settings must be retrieved from environment variables (for example, Robot Framework com-port)
3. Add the required environment variables to the Node settings.
4. Install and configure robot framework plugin to see the test status.
5. Add a stage that is parallel with build stage. This stage runs Robocop linter on your RobotFramework files, produces SARIF-report and parses the result to Jenkins report.

The idea is to make building and testing scripts independent of the build machine. For example, we could have two nodes with different serial port numbers/names and openocd installation directory and still be able to build and test on either node with the same pipeline.

You need to have two plugins installed:

- Robot Framework plugin
- Warnings plugin for reading Sarif report

<https://github.com/MarketSquare/robotframework-robocop/blob/main/docs/integrations/jenkins.md>

https://github.com/MarketSquare/robotframework-robocop/blob/main/docs/configuration/configuration_reference.md

Show a demo of the pipeline activation to the instructor:

1. Your project folder and all scripts
2. Git hook configuration.
3. Git hook script file.
4. Your node's local variable configuration.
5. Pipeline activity that shows Robot Framework status and commit messages from several commits.
6. Do a commit to trigger tests.

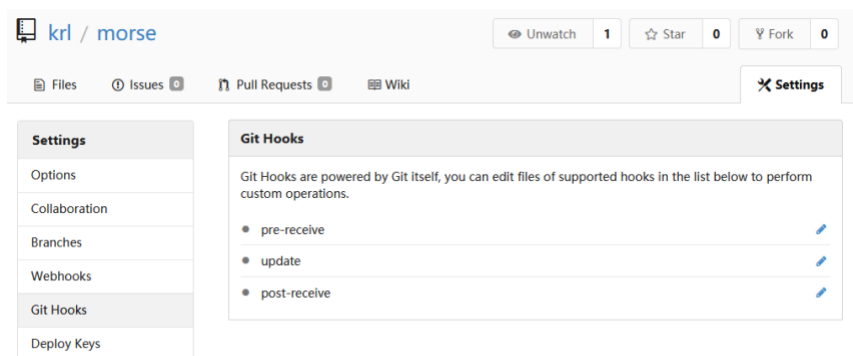
Build triggers

The following article explains the principle triggering builds externally. It can't be directly applied to current version of Jenkins, but you can find the differences when you set "Trigger builds remotely" in the pipeline configuration.

<https://kohsuke.org/2011/12/01/polling-must-die-triggering-jenkins-builds-from-a-git-hook/>

Note: Our server address is: <http://jenkins:8080/>...

You can use curl in gogs scripts to fetch (=trigger) from an URL. Make sure that branch name matches your main branch in the post-receive trigger.



You need to enable anonymous read access in Jenkins Security for the trigger to work. There is also a less global role based setting but that requires extensive configuration and does not provide added value for our test setup.

Authorization

Logged-in users can do anything

☒ Allow anonymous read access ?

You can see if build is triggered but making a small change to a source file for example, edit a comment, and commit the change to repository. Repository scanning should start practically immediately and then the Pipeline is run.

Environment variables

Per node environment variables are added in node configuration. Setting appear when you activate environment variables in Node properties.

Robot Framework plugin

When the plugin is installed, a step to configure the plugin is added to snippet generator:

Sample Step

robot: Configure robot framework report collection